

Brainstorming Ideas and Process

Below are several sketches generated by our team members during our collaborative brainstorming session

Idea 1 (Log-in page): This is a sketch of our prospective log in page that we generated. Users should be able to choose to log in and sign up pretty easily and the ability to log in with either Facebook, Google or Apple should be provided. This is a pretty basic idea, and so only one sketch was drawn up.



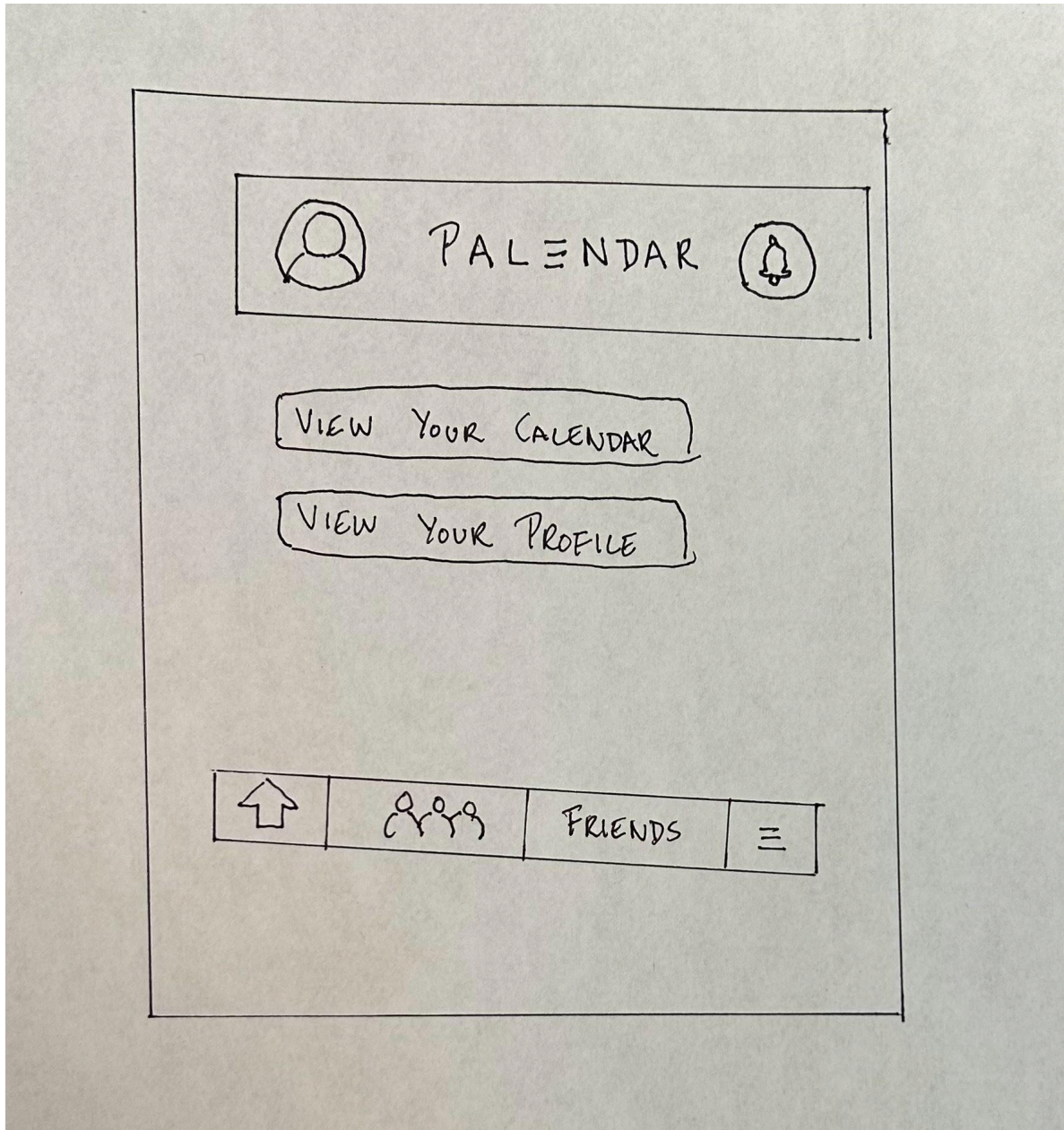
Sketch by Sahil Sanghvi

Idea 2 (Prospective Home Page 1): This is a sketch of the first idea we had for the landing page. It directs you to your own personal calendar and there is a navigation bar at the bottom that allows you to view your homepage, your groups and your friends. There is also an option to view certain additional pages corresponding to should be met/could be met design requirements that may be implemented.



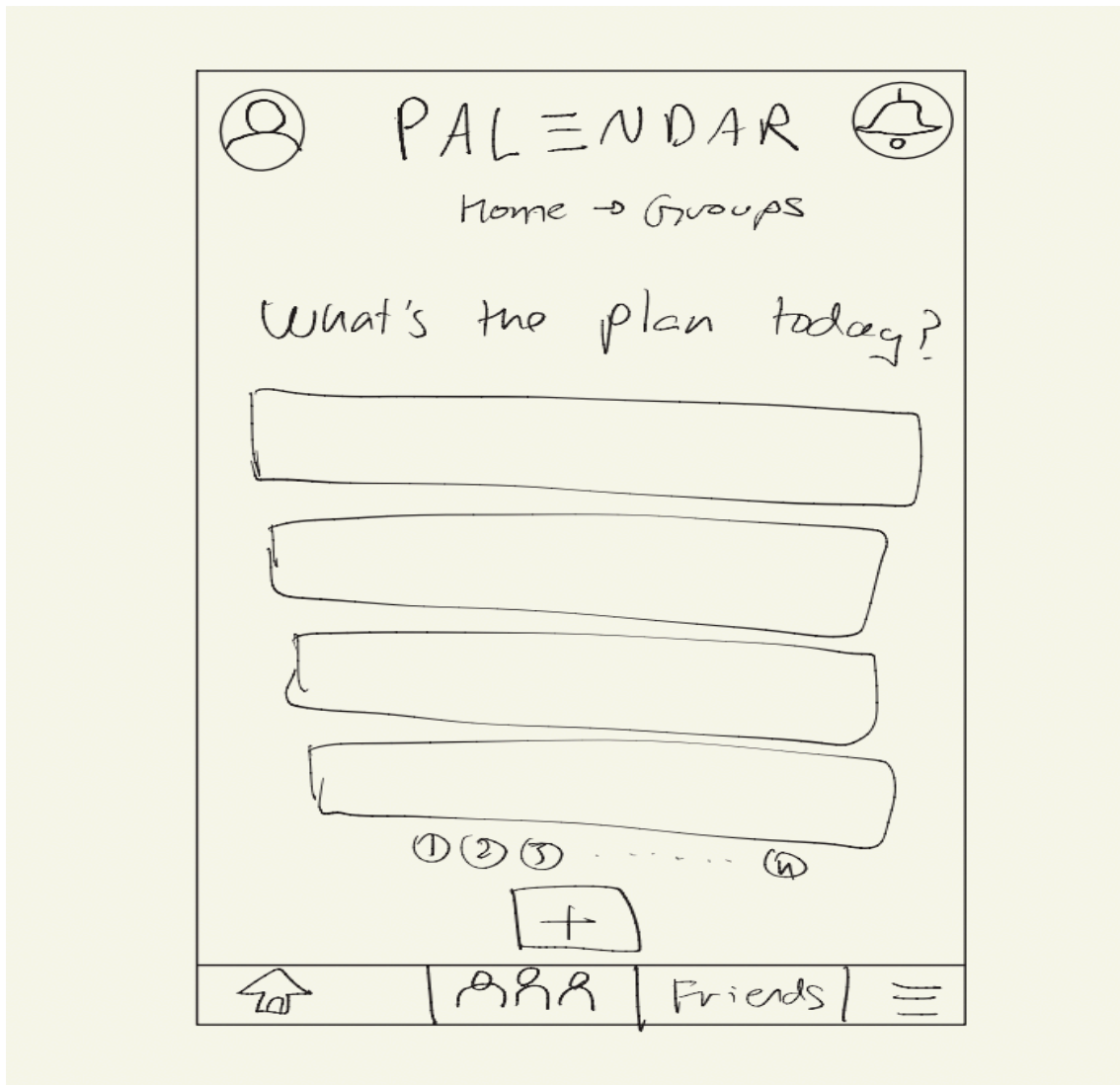
Sketch by Sahil Sanghvi

Idea 3 (Prospective Home Page 2): This sketch represents another idea we had regarding the home page. Instead of directly landing on their calendar, here, the user will be greeted with two buttons. One to view and customize their personal calendar and another to view their profile details. This idea was implemented in order to involve a separation of concerns based approach.



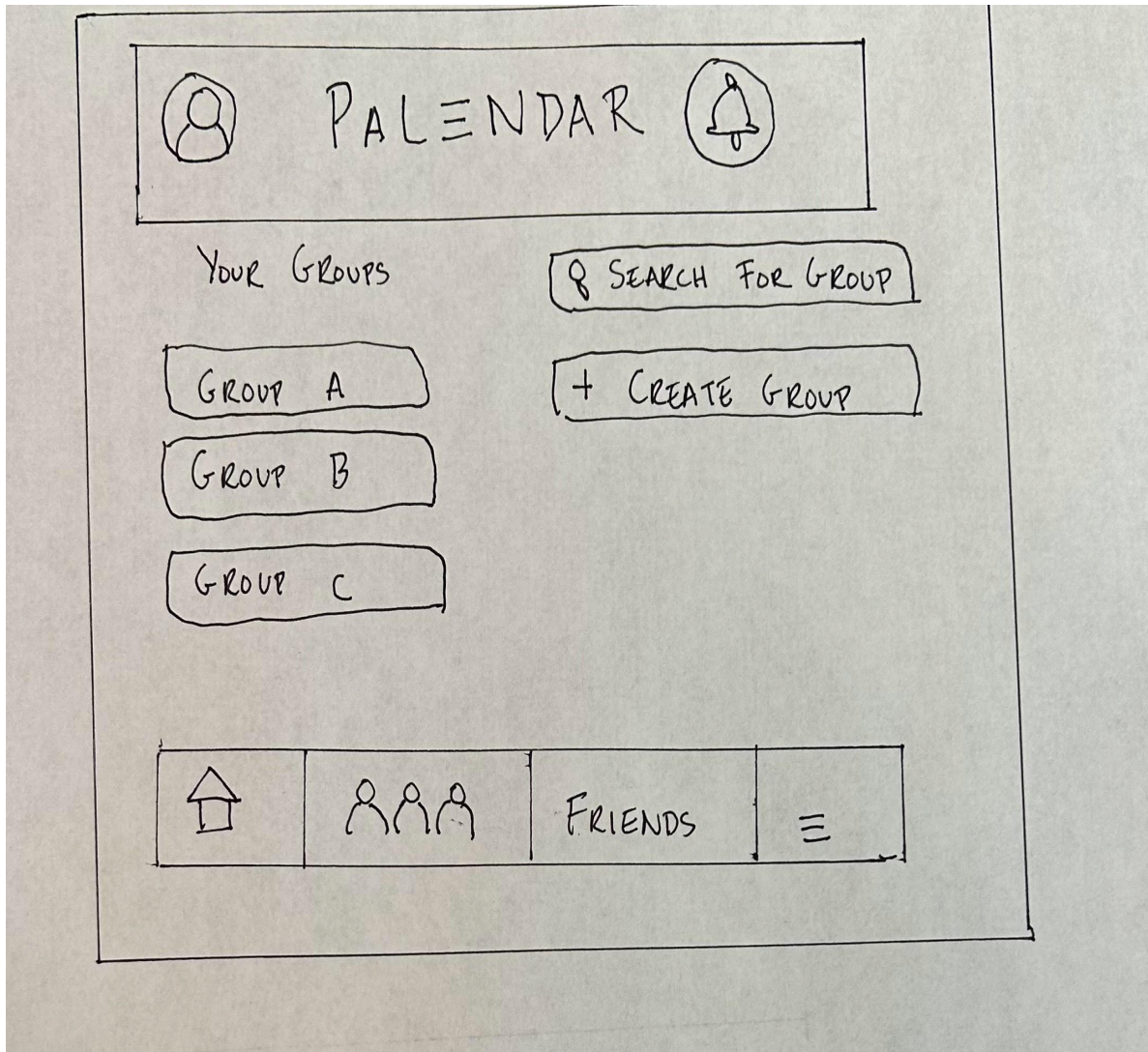
Sketch by Rishon Rajesh

Idea 4 (Prospective Group Landing Page 1): This is the first idea we had for the group landing page we had which involves the user being able to view the multiple groups they are a part of and offers them the ability to click on the group to view the group calendar. There is also a plus button which the user can click on in order to create a new group or search for another group to join.



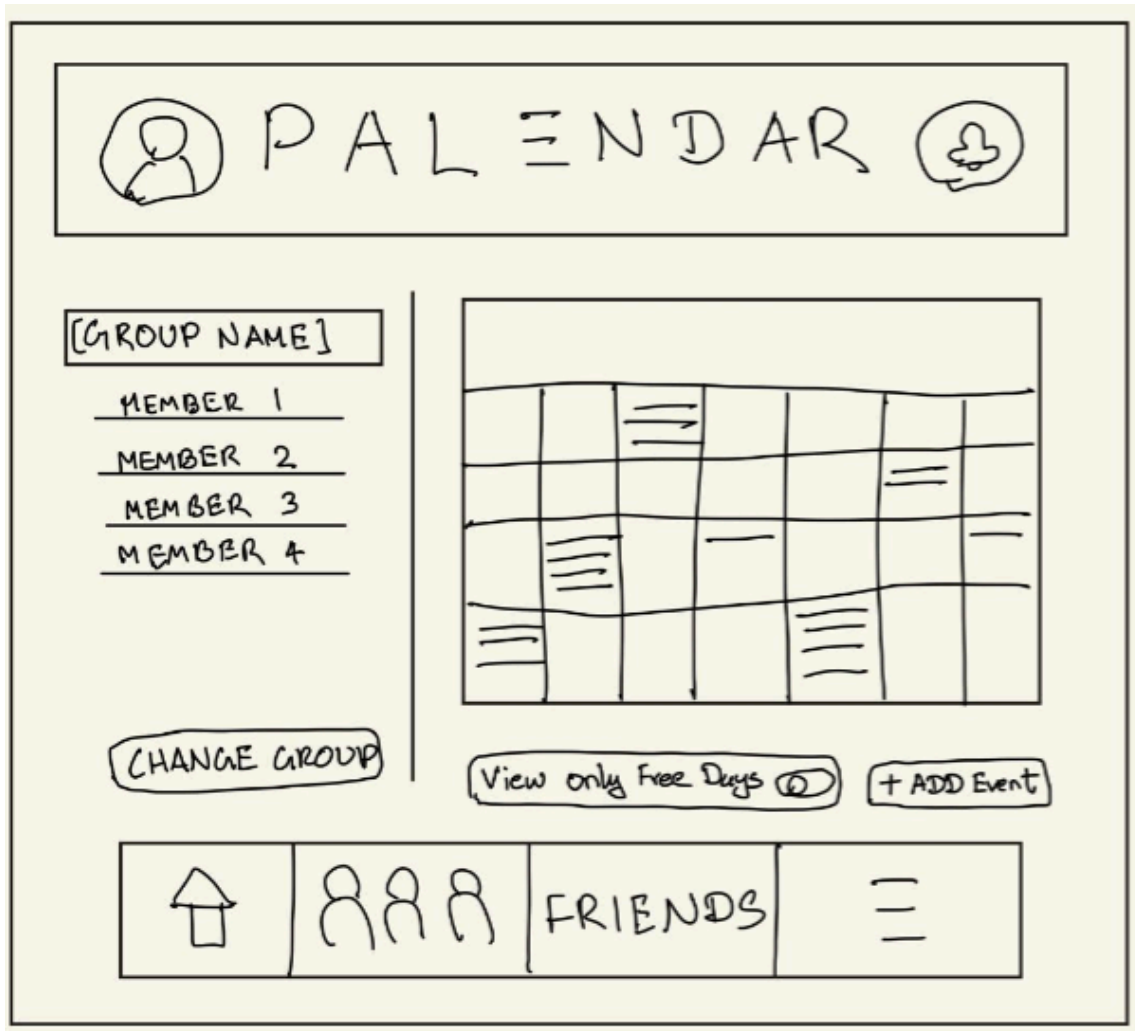
Sketch by Sahil Sanghvi

Idea 5 (Prospective Group Landing Page 2) : This sketch involves another approach to the group page. This is very similar to the first sketch except for two key differences. The first difference is a deviation from the page based group layout to a scrolling one, where users will be able to scroll down to view all their different groups instead of moving page by page. The second is the removal of the plus button in order to add two separate buttons to create or search for a group.



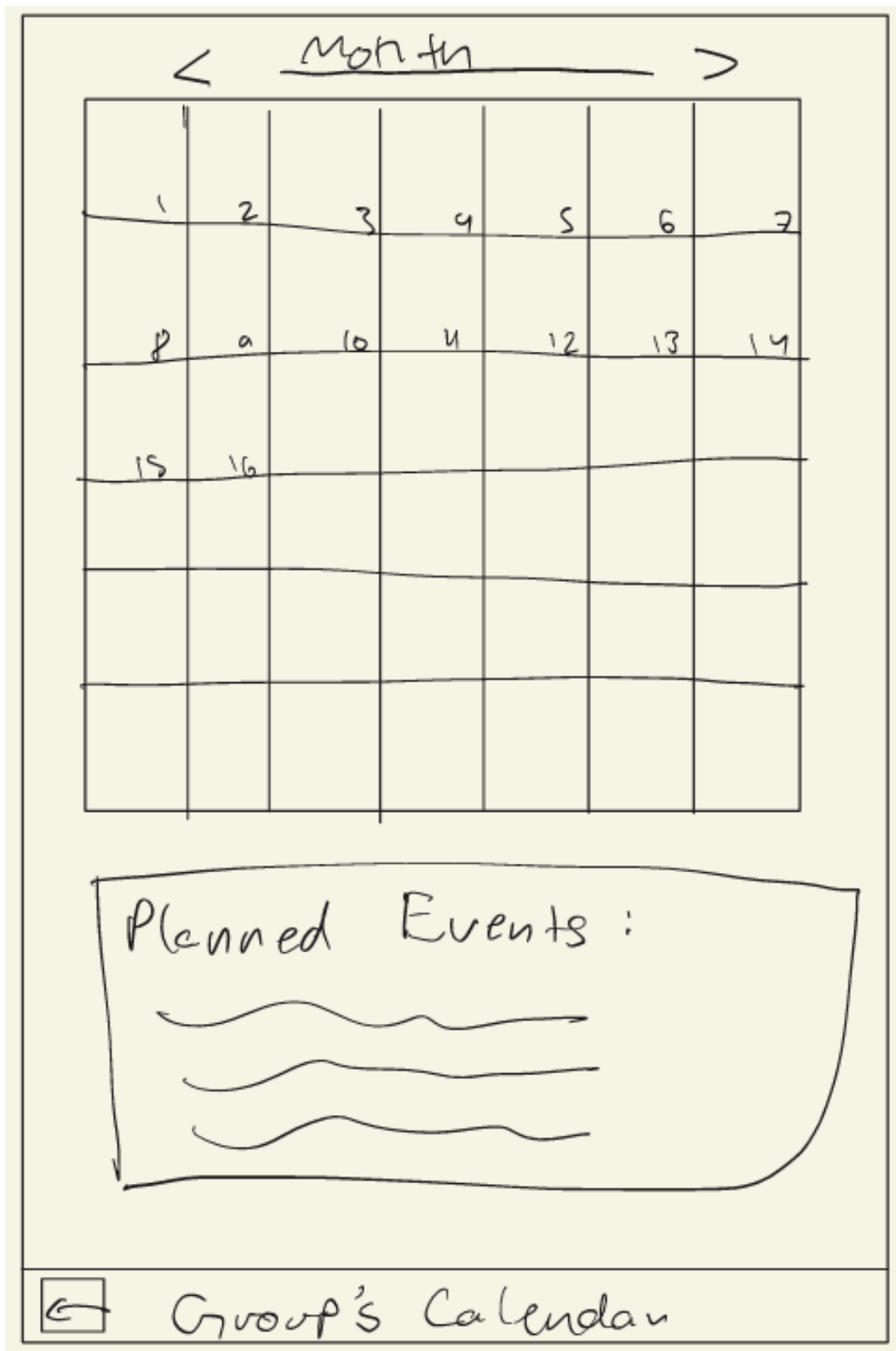
Sketch by Rishon Rajesh

Idea 6 (Prospective Group Calendar Page 1) : This is a sketch of the original group calendar page that we developed. It involves a lot more features than our second sketch and takes an all encompassing approach. It gives the user a clear visual representation of all the operations they can do with the group calendar.



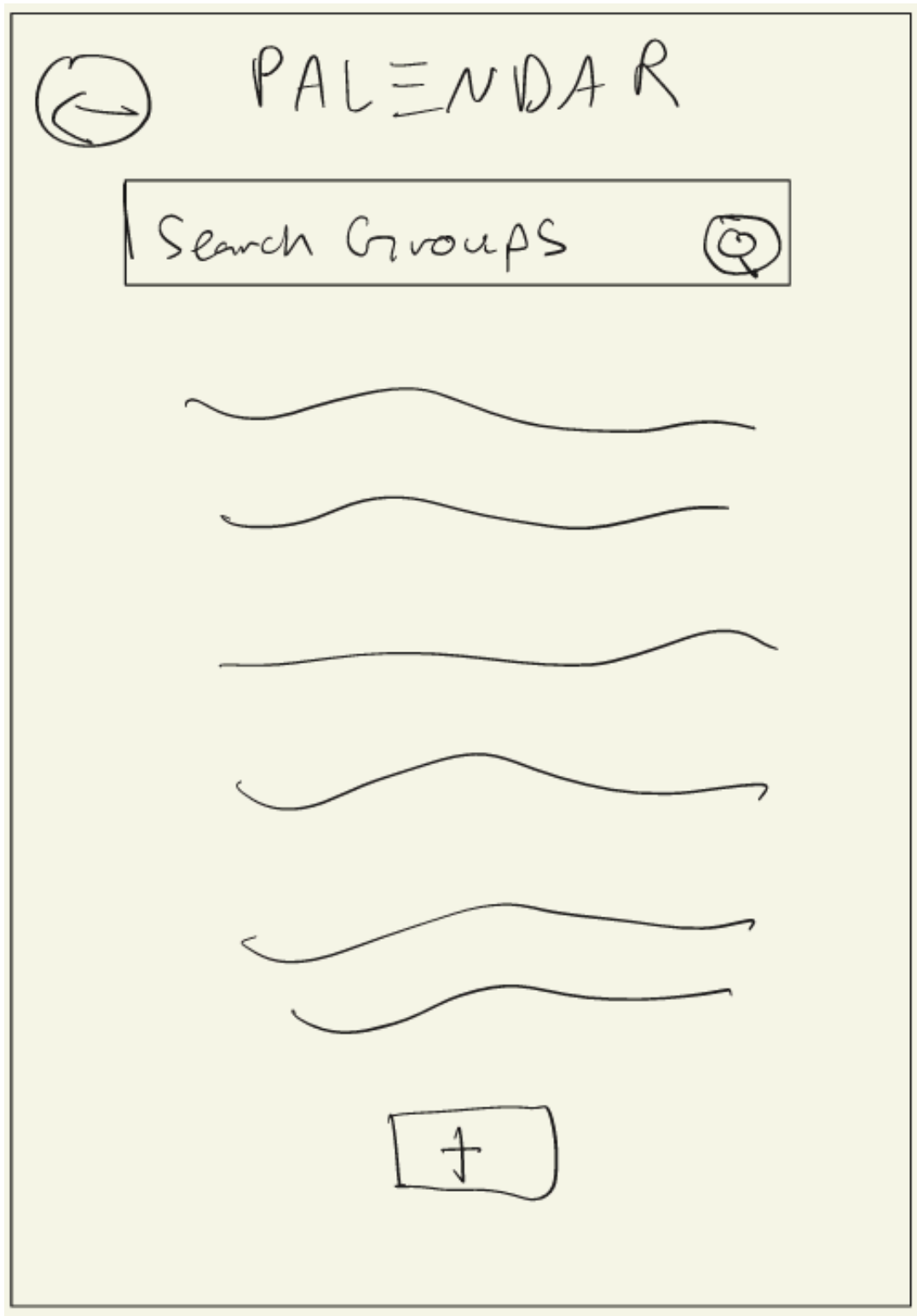
Sketch by Sam Sanjay Satish

Idea 7 (Prospective Group Calendar Page 2) : This is a sketch of the prospective group calendar page with a less cluttered user interface. Here, there is a centered focus on the calendar and planned events itself rather than a lot of options that may confuse the user.



Sketch by Manya Juneja

Idea 8 (Prospective Group Search Page 1) : Below is the group search page we had initially designed which allows users to search for a group and view all possible groups with a similar name.



Sketch by Manya Juneja

Idea 9 (Prospective Group Search Page 2) : While implementing our second sketch, we noticed that anonymity was an important concern and that a user probably should not be able to view all random groups with codes similar to the one they wanted to join. So, we ended up on this design which directly translates into our final one



Sketch by Sahil Sanghvi

Idea 10 (Prospective Create Group Page) : This sketch depicts the create group page which the user will enter when they want to create a new group for their friend group. Here they will be asked to enter a group name, select a category for the group and enter a description. Then they can click on the generate code button to create the nine digit code and group. We have only included one sketch here as this is a relatively straightforward feature.

PALENDAR

CREATE GROUP

ENTER GROUP NAME:

SELECT GROUP TYPE:

FRIENDS

FAMILY

CLUB

OTHER...

ENTER GROUP DESCRIPTION:

GENERATE CODE

Navigation bar: FRIENDS

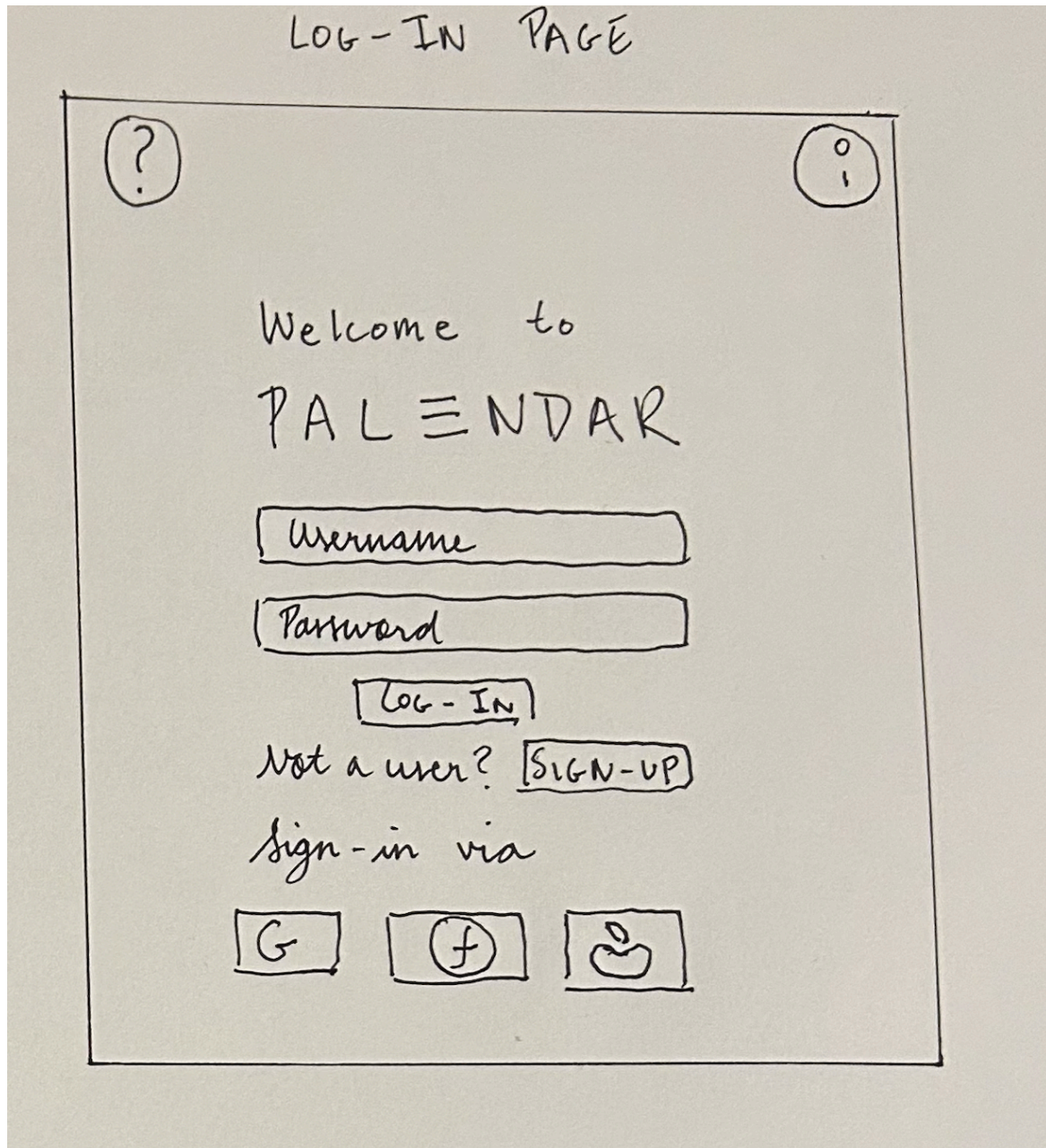
Sketch by Rishon Rajesh

Analysis of Ideas and Idea Selection + Final Sketches

After the brainstorming process, we evaluated and analysed our sketches in order to decide which features to implement in our sketch of the finalised idea for each component. We relied on two main evaluation strategies, voting and the four categories method, in order to select the features that go into our final sketch. We also used two academic articles to combat our key ideological debate during the brainstorm: the idea of aesthetics vs usability. This is portrayed especially in our final sketch of the group calendar page and create group page. The articles we used for this are: Tuch, A. N., Roth, S. P., Hornbæk, K., Opwis, K., & Vargas-Avila, J. A. (2012). Is beautiful really usable? Toward understanding the relation between usability, aesthetics, and affect in HCI. *Computers in Human Behavior*, 28(5), 1596–1607 and Hassenzahl, Marc. (2004). The Interplay of Beauty, Goodness, and Usability in Interactive Products. *Human-Computer Interaction*. 19. 319–349. 10.1207/s15327051hci1904_2.

Component 1 (Log-in page)

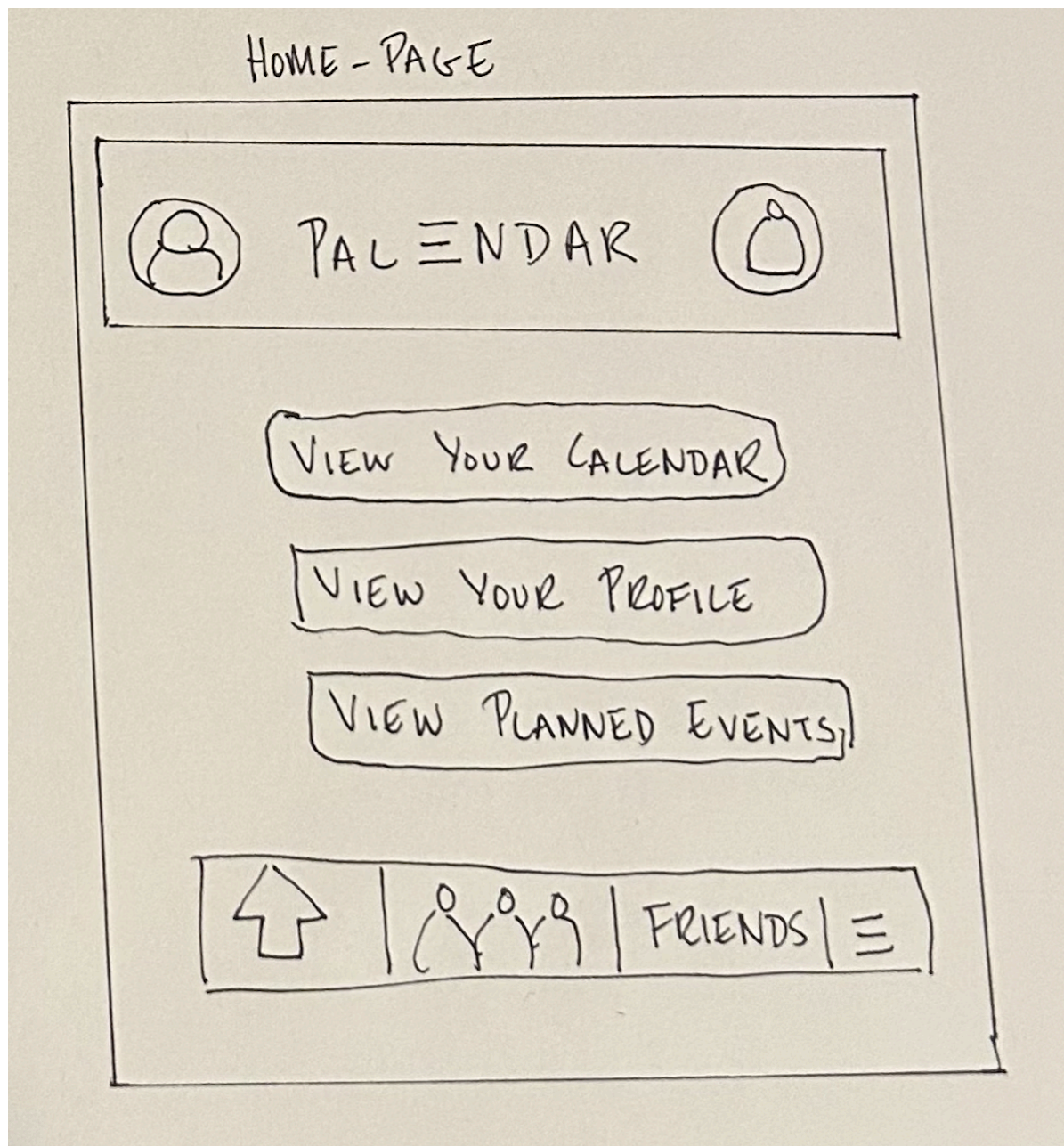
We were pretty happy with our initial brainstorm sketch for the Log in/Sign up page as it is a relatively straightforward feature. The Log-in/Sign-up form and accompanying features all made it into the most rational features category. The sign in via Google, Facebook or Apple option was also a part of the most rational category but we also thought that it might be considered as a Darling category feature due to the ease of access that it provides the user.



Sketch by Rishon Rajesh

Component 2 (Home page)

For the home page, we had relatively different ideas in our two brainstorm sketches, and we had to put this to a vote. In particular, we voted on whether it would involve a calendar in the main interface or whether we would go along with the separation of concerns approach explained in the second brainstorm sketch of this feature. After voting, we decided to go with the separation of concerns approach alongside a new third button that will allow the user to access and view all their planned events. This is where the user will end up upon logging in.

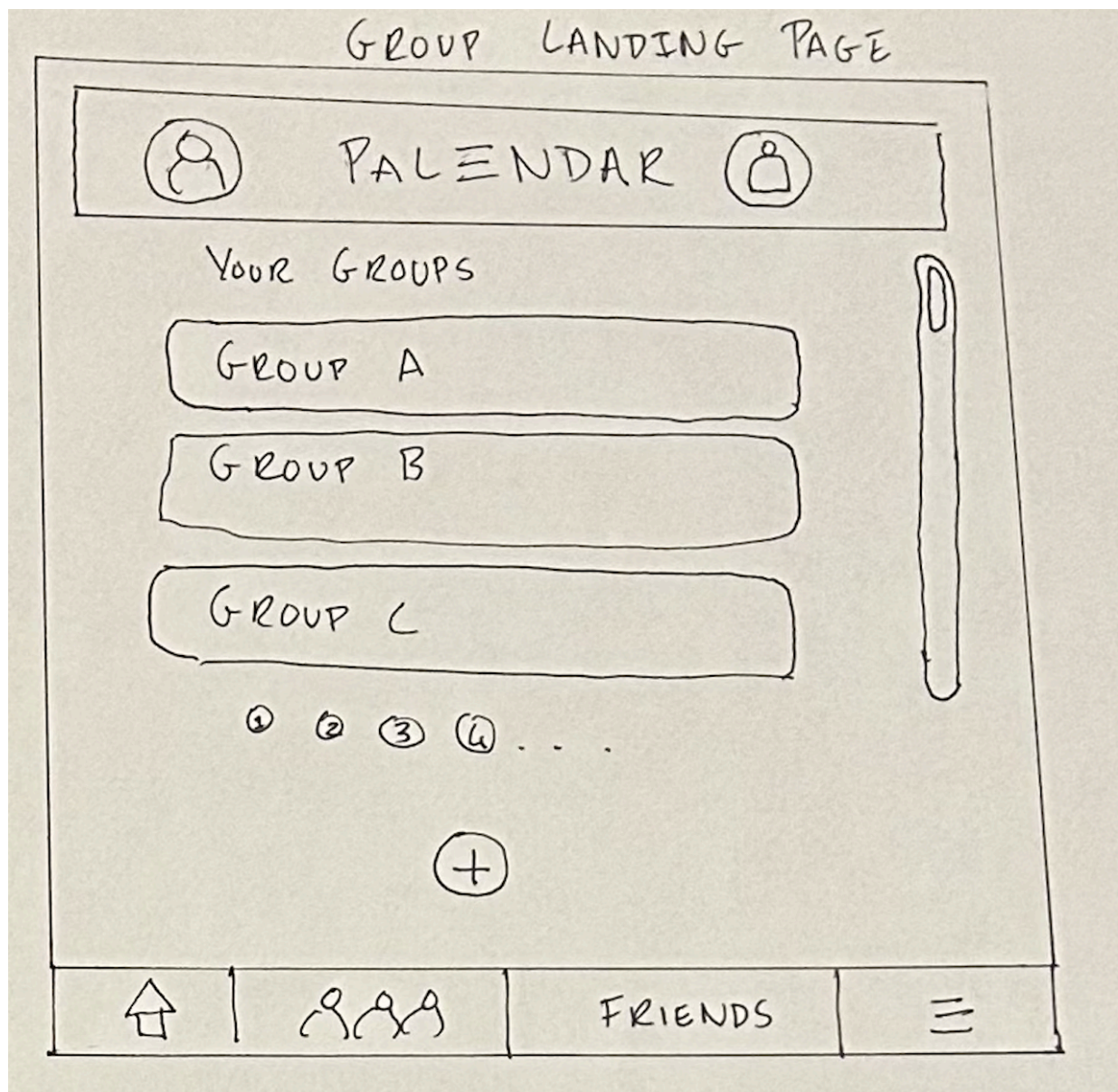


Sketch by Rishon Rajesh

Component 3 (Group Landing page)

The group landing page is one of our most important features. The main debate we had in this part was as to whether we include a plus button at the bottom of this page in order to allow the user to create or search for a group or move forward with a straight up button and textbox based interface. Here, we used the four categories method in order to categorize our ideas and decide upon the final one. As the plus button added creative flair to the web page as well, it was in the Darling category and upon voting, this was the implementation we decided to proceed with.

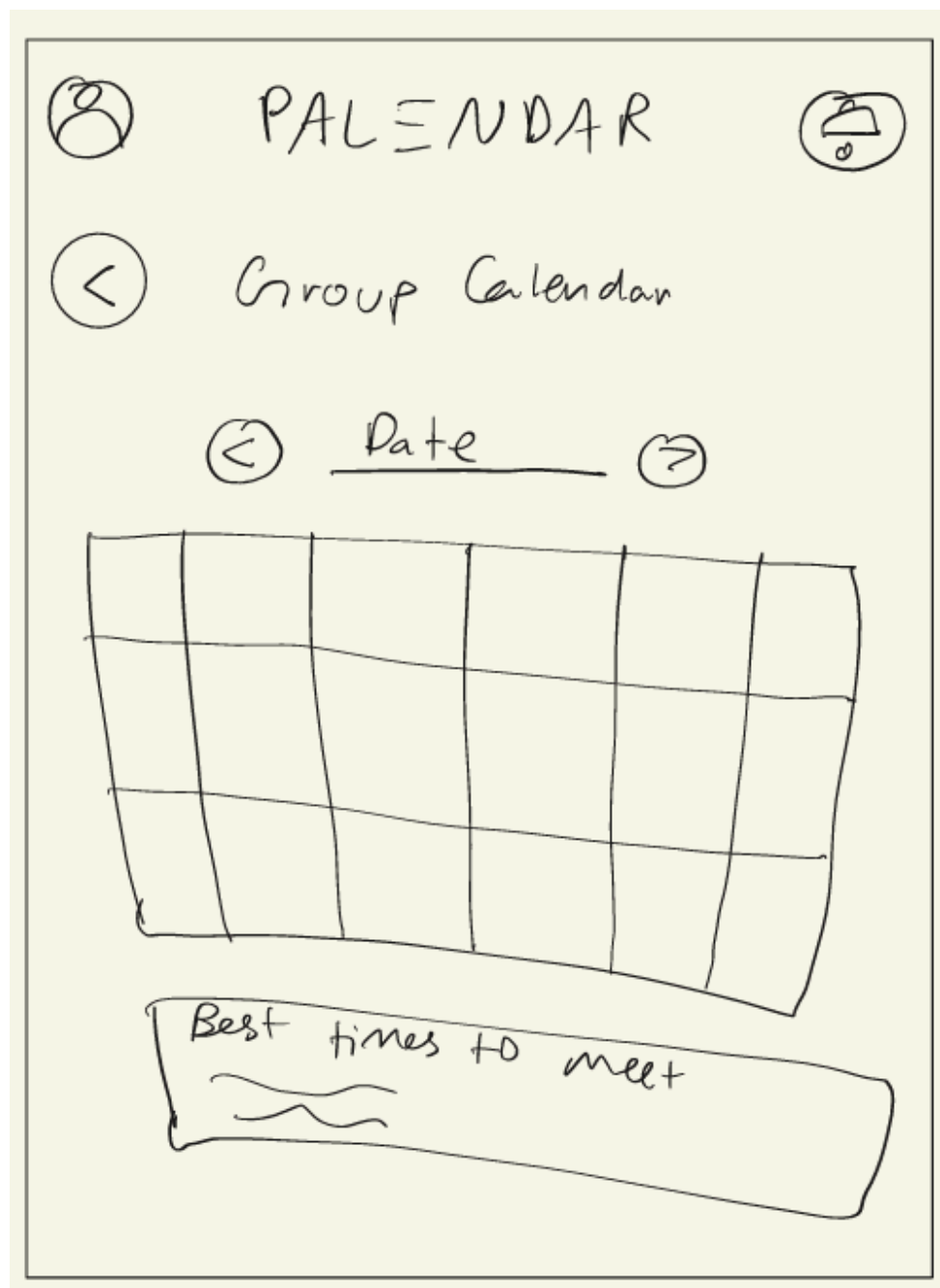
We also had to vote on whether we used a scrolling based or page based interface for the groups being displayed but decided that grouping both together and moving forward with a combination of both would provide the best user experience.



Sketch by Rishon Rajesh

Component 4 (Group Calendar page)

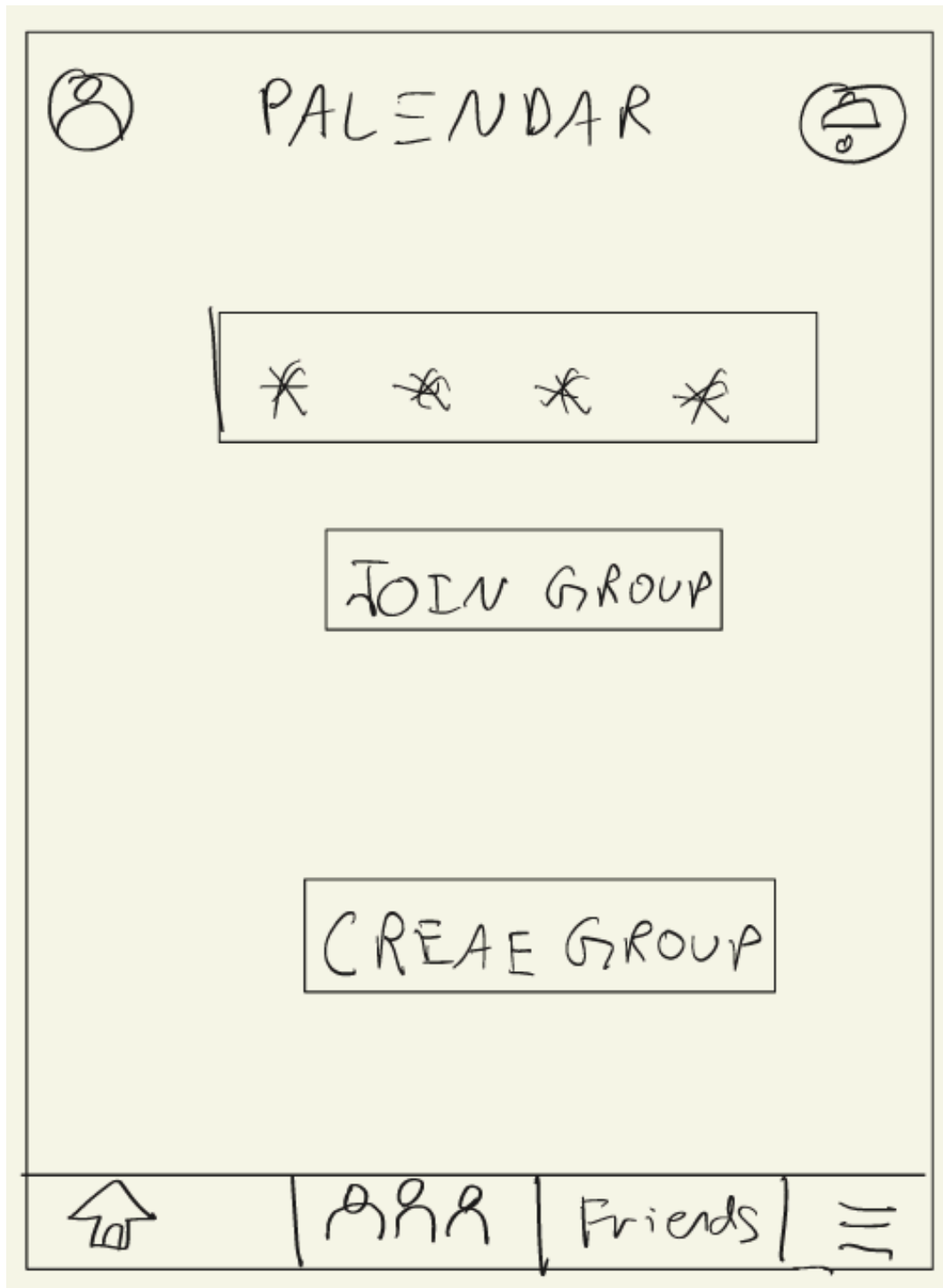
For our finalised group calendar page, we stuck to a combination of voting and qualitative discussion in order to decide our last approach. We thought the initial brainstorm sketch had too many buttons and random features that made the UI cluttered and messy to see. So, we decided on implementing only the key features: the group calendar itself, and a depiction of the best times that the group can meet. Also as a possible darling category feature, we chose adding colour coding to different hours of the week as suggested by Abhee in her user interview.



Sketch by Sahil Sanghvi

Component 5 (Group Search page)

The anonymity concerns that were discussed after our first brainstorm sketch directly fed into the finalized sketch and the desire to keep groups anonymous proceeded with a unanimous vote. Here, we stuck with a simple layout in order to avoid any clutter.



Sketch by Sahil Sanghvi

Component 6 (Create group page)

This is also a relatively straightforward component and we didn't have to decide between too many features so there was not much to do with voting. However, when we did our analysis of grouping ideas into the four categories, we really enjoyed the idea of group categories and creative icons. This insight added itself to the Darling category. Since it is relatively easy to implement, we are also moving forward with this. Besides this we wanted to add a create group message before the generate code button in order to make the interface seem more straight forward.

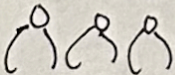
CREATE GROUP PAGE

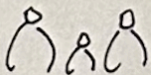
 PALINDAR 

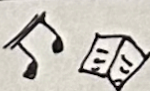
CREATE GROUP

ENTER GROUP NAME:

SELECT GROUP TYPE


FRIENDS


FAMILY


CLUB

ENTER GROUP DESCRIPTION:

CREATE GROUP:

Sketch by Rishon Rajesh

Most Promising Idea

Our most promising idea that we spent the most time analysing and voting on was definitely that of the group calendar. We really put effort into voting, grouping into four categories and coming up with features such as colour coding. We heavily debated the usability vs. aesthetics of this and came up with the most optimised approach upon reviewing both our referenced articles.

Storyboarding

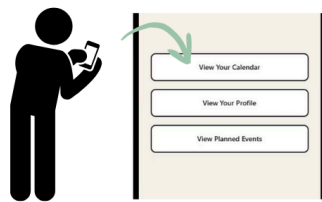
HANGING OUT WITH FRIENDS USING PALENDAR

Planning a Hangout



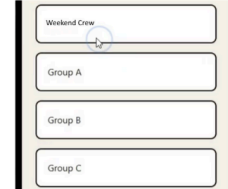
The user decides they want to hang out with their friends from an existing Palendar group. They open the Palendar app on their phone to start planning.

Opening the Home Page



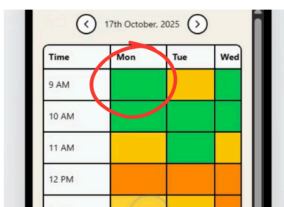
The Home page loads, displaying options such as View Calendar, View Profile, and Planned Events. The user taps View Calendar to begin organizing the hangout.

Selecting the Friend Group



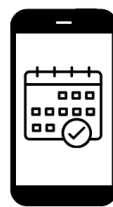
On the Group Landing page, the user sees a list of groups they are part of (e.g., "Weekend Crew"). They tap the group they want to hang out with to view the shared calendar.

Finding a Common Time



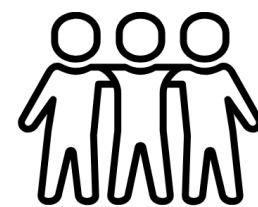
The Group Calendar shows everyone's availability in color-coded slots (green = available, yellow = limited, red = unavailable). The user finds a green time slot that works for everyone.

Creating the Event



The user selects the green slot and clicks Create Event. A confirmation message appears saying, "Event Scheduled Successfully!" The event is now visible under Planned Events.

Hanging Out Together



On the scheduled day, the group meets up successfully. Palendar helped them quickly find a mutual time and plan the hangout with minimal effort.

Heuristic Evaluation

Process

Our team conducted a heuristic evaluation of our final Palendar interface using Jakob Nielsen's ten usability heuristics covered in class. Each member independently reviewed all six interface components : Log-in, Home, Group Landing, Join/Create Group, and Group Calendar. We then met as a group to combine our findings, remove duplicates, and agree on final severity ratings using the 0–4 scale (0 = not a problem, 4 = usability catastrophe).

This process helped us systematically inspect our design without running a full usability study, while still capturing potential problems early in the prototype stage.

Summary of Findings

Overall, the evaluation showed that the Palendar interface follows several strong usability practices such as consistency, simple language, and minimalist layout. Most issues were minor and related to visibility of system status, recognition rather than recall, and aesthetic and minimalist design.

The Group Calendar and Join/Create Group pages contained the most notable issues, mainly around missing feedback and unclear user flow. None of the problems were severe enough to prevent task completion, but addressing them would significantly improve clarity and efficiency for first-time users.

Heuristic Evaluation Table

Component / Page	Heuristic Violated	Problem Description	Severity (0–4)	Suggested Fix
Log-in Page	1. Visibility of System Status	No visual confirmation after pressing “Log-In”	3 (Major)	Add a brief loading spinner.
	2. Match Between System and Real World	“Username” label may confuse users expecting “Email.”	2 (Minor)	Change field label to “Email Address.”
Home Page	4. Consistency and Standards	Button text (“View Your Calendar,” etc.) lacks visual hierarchy.	1 (Cosmetic)	Make the main button slightly larger or highlighted.
	7. Recognition Rather Than Recall	Navigation icons at the bottom are minimal; unclear which page is active.	2 (Minor)	Add highlight or underline to show the active tab.
Group Landing Page	8. Aesthetic and Minimalist Design	Scrolling dots and plus buttons may not be intuitive for all users.	2 (Minor)	Label “+ Create/Join Group” below the plus icon for clarity.
Join/Create Group Page	5. Help Users Recognize, Diagnose and Recover From Errors	No example or hint for code format.	2 (Minor)	Add placeholder text (e.g., “Enter 9-digit Group Code”).
Group Calendar Page	6. Error Prevention	No confirmation before editing/deleting calendar entries (if interactive).	3 (Major)	Add confirmation dialog before changes.
	8. Aesthetic and Minimalist Design	Dense color blocks can appear cluttered on small screens.	2 (Minor)	Simplify color palette or add toggle for compact view.

Reflection

The heuristic evaluation confirmed that our team's focus on simplicity and clarity has resulted in a mostly usable design, though certain aspects of feedback, error handling, and information visibility need improvement.

This activity also reinforced the balance between aesthetic appeal and usability that we explored during our brainstorming and article analysis phase. Implementing the suggested fixes will make Palendar more intuitive, particularly for first-time users, while maintaining the minimalist and accessible design we aimed for throughout the project.

References

Hassenzahl, Marc. (2004). The Interplay of Beauty, Goodness, and Usability in Interactive Products. *Human-Computer Interaction*. 19. 319–349. 10.1207/s15327051hci1904_2.

Tuch, A. N., Roth, S. P., Hornbæk, K., Opwis, K., & Bargas-Avila, J. A. (2012). Is beautiful really usable? Toward understanding the relation between usability, aesthetics, and affect in HCI. *Computers in Human Behavior*, 28(5), 1596–1607.
<https://doi.org/10.1016/j.chb.2012.03.024>